

Ideas:



Categorize:

1. Trend Panel
 - Line chart
2. Comparison Panel
 - bar chart
3. Pie chart.



source: different
consumers different

Combine & refine:

National trends:

↳ Line + Timeline



↳ State comparisons

Proportional symbols.

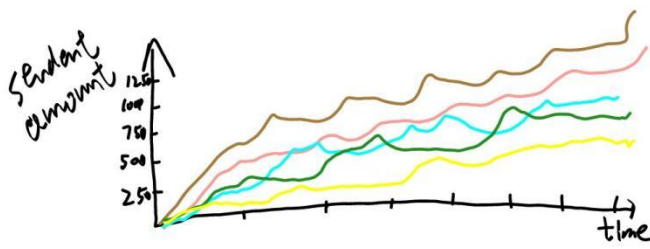
↳ Proportional

↳ Heat map.

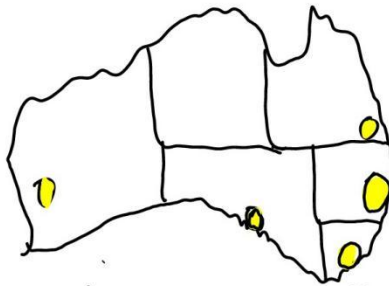
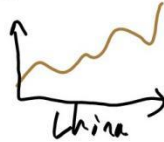
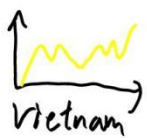
Question:

- Does it meet the requirements of the input data?
eg. size < 1MB.
- Did you answer the research question?
- Is it easy for the audience to understand?

Layout:



China
India
Nepal
Brazil
Vietnam



Proportional symbol map



Title: National Trend

Author: Zhanghang Zhang

Date: 15/10/2025

Sheets: 2

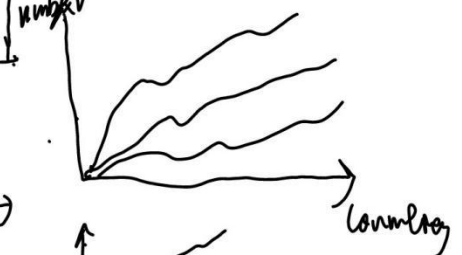
Task: Design a trend chart.

operation:



top 3 total
China
India
Nepal

Click number



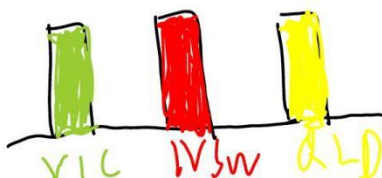
Top 3 Total

Focus:

Core idea:

Regional distribution comparison

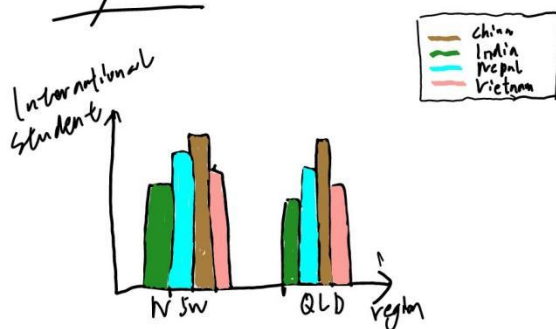
- compare international student numbers by state.
- show the main difference in the distribution of source countries.
- The visual focus is on two states of NSW and VIC.



Discussion:

- Does the map and bar combination clearly show state differences?
- Is colour encoding intuitive for country comparison?
- Does the tooltip help users understand the data faster?

Layout:



use toggle button to switch between Grouped and Stacked view.

stacked bar chart
(proportion of international student in each state)

Title: State Comparison

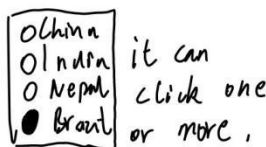
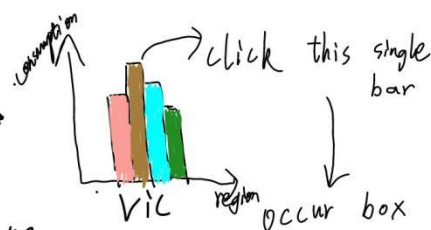
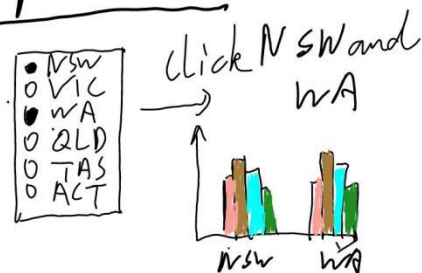
Author: Zhonghang Zhang

Date: 15/10/2025

Sheets: 3

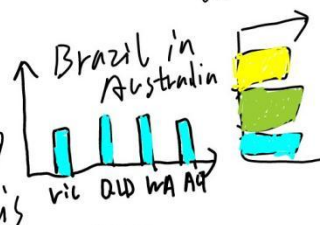
Task: Design a different region chart

operation:



it can click one or more.

Click and appear this



Focus:

Core Idea:

structural comparison Across states.

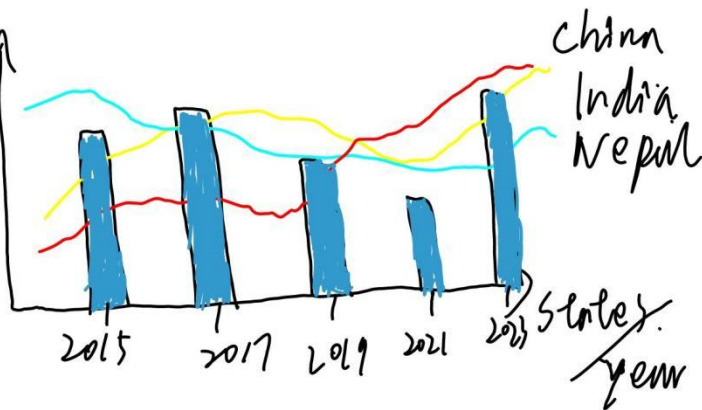
- NSW & VIC → high proportional from China.
- QLD → rising share from India.
- Smaller states → greater share from Nepal.

Discussion:

- Does switching between grouped and stacked help show structure vs. quantity?
- Is the color scheme sufficient for distinguishing countries?
- Does the chart connect with the map?

Layout:

Student membership



Mark significant events too.

↓ drop and recover. ←
2019
- min timeline

Title: Student vs. time

Author: Zhongheng Zhang

Date: 15/10/2025

Sheets: 4

Task: Design a multi chart

operation:

2015 2025

Hover →



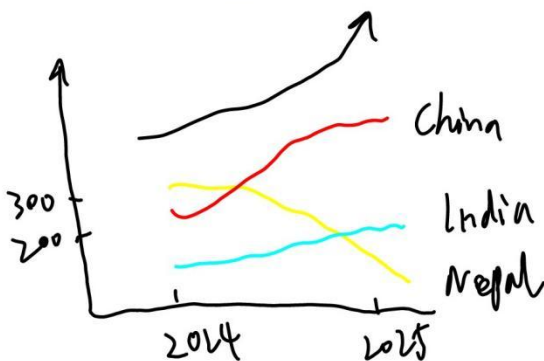
Click → China



Focus:

Core concept:

Trend comparison over time.



→ min timeline.

Discussion:

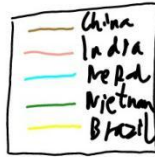
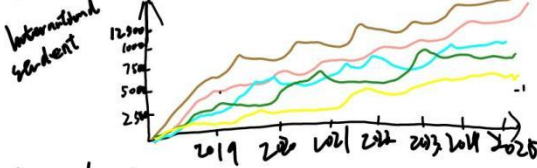
- Does the line chart clearly show the change over time?
- Are the slope graphs effective for yearly comparison?
- Is the hover/click interaction intuitive?

Layout:

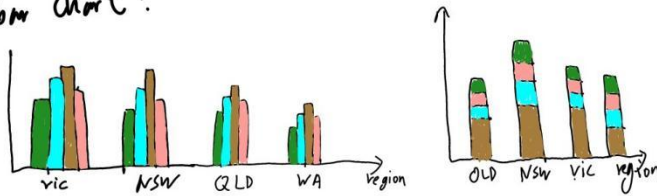
Proportional map:



Line chart:



bar chart:



Title: Final Realization

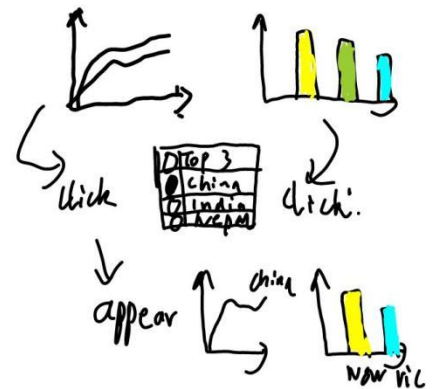
Author: Zhonghang Zhang

Date: 15/10/2025

Sheets: 5

Task: Design the final dashboard.

operation:



Focus:

Core idea:

Integrated overview + interaction

Combine space, structure, time
3D dimensional information of

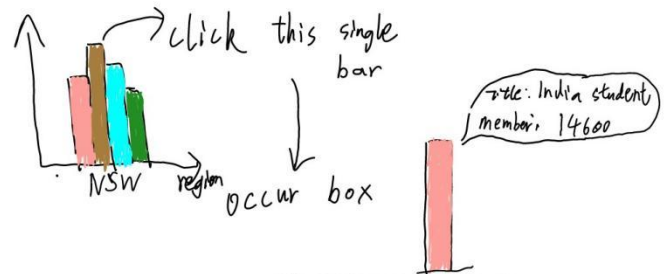
→ which state has most students which country leads?

→ high grows rapidly.

• NSW, VIC lead

• China recovers after pandemic

• India grows steadily.



Detail:

Date Sources: 1. Australian Government Department of Education
2. GeoSON - Australian States.

Implement tool: 1. Vega-Lite + JSON + HTML
2. Interactive binding with params for dropdown / hover / click

Development step:

1. Data cleaning & csv merge.
2. Base layout (Map + Bar + Line)
3. Add dropdown + toggle buttons
4. Add tooltip + hover interactivity.
5. Adjust layout responsiveness (LMB).